

Chapter3: Multiplexer, Latch, Buffer

Asst.Prof.Dr.Supakit Nootyaskool



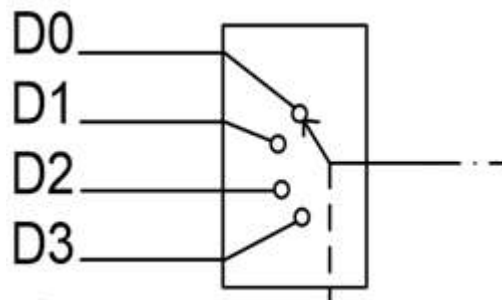
Objective

- Describe concept multiplexer and de-multiplexer circuit.
- Compare and contrast latch circuit and buffer circuit.
- Explain truth table of set-reset flipflop.

Topic

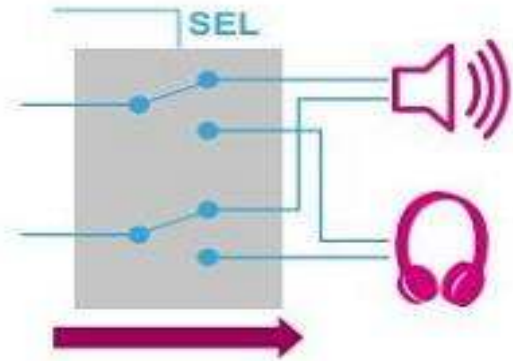
- Multiplexer circuit
- Demultiplexer circuit
- Latch circuit
- Buffer circuit
- Set-Reset Flipflop

Selector switch

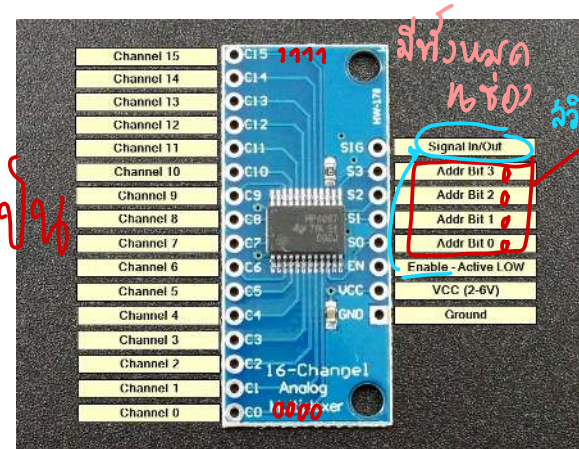


Selector switch is mechanical switch for selecting the signal channels
We found the selector switch in electrical machines, radio event if the car.

Switch -> Multiplexer (MUX)



นพ ๗๗
จดเอาไว้ไม่เสีย



LazGlobal CD74HC4067 16-Channel Analog Digital Multiplexer Breakout Board Module For Arduino

★★★★★ 1 คะแนน

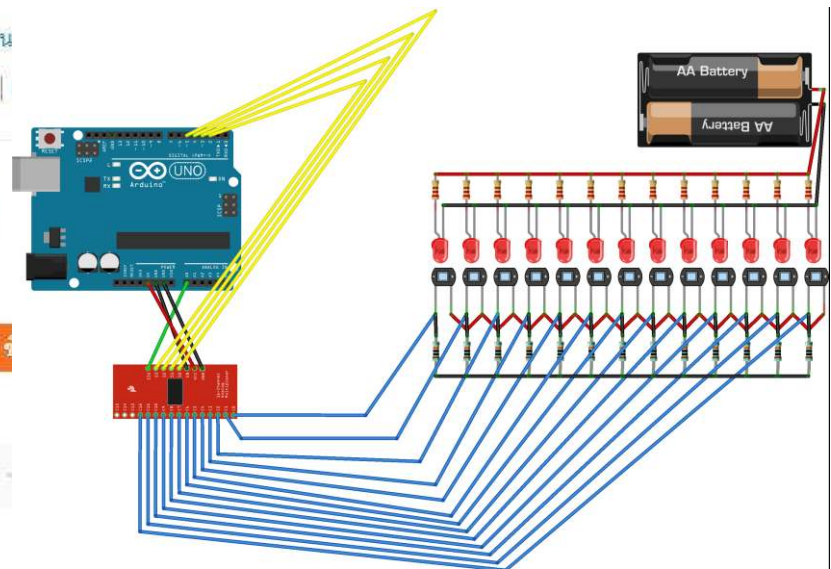
แบรนด์: No Brand

฿36.00

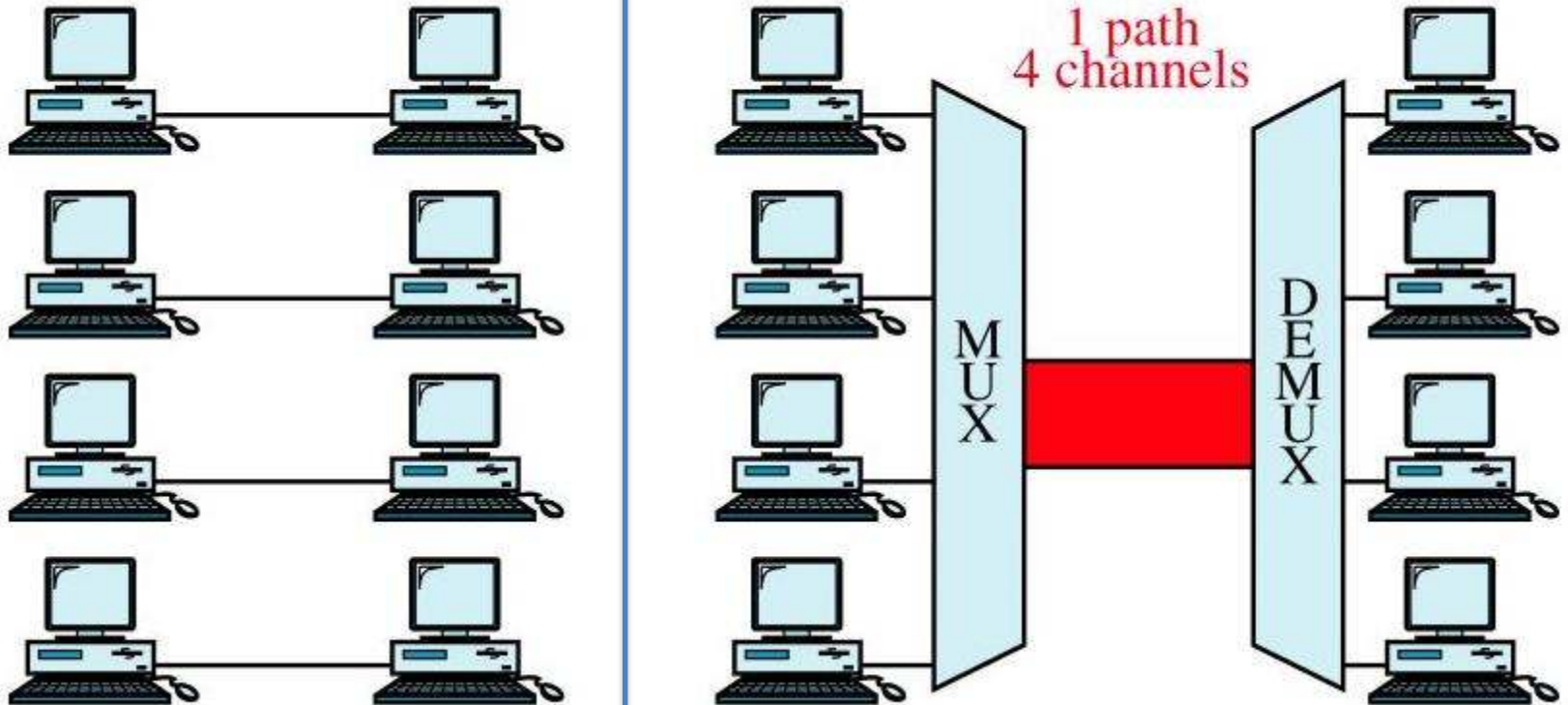
฿50.00 -28%

โปรโมชั่น

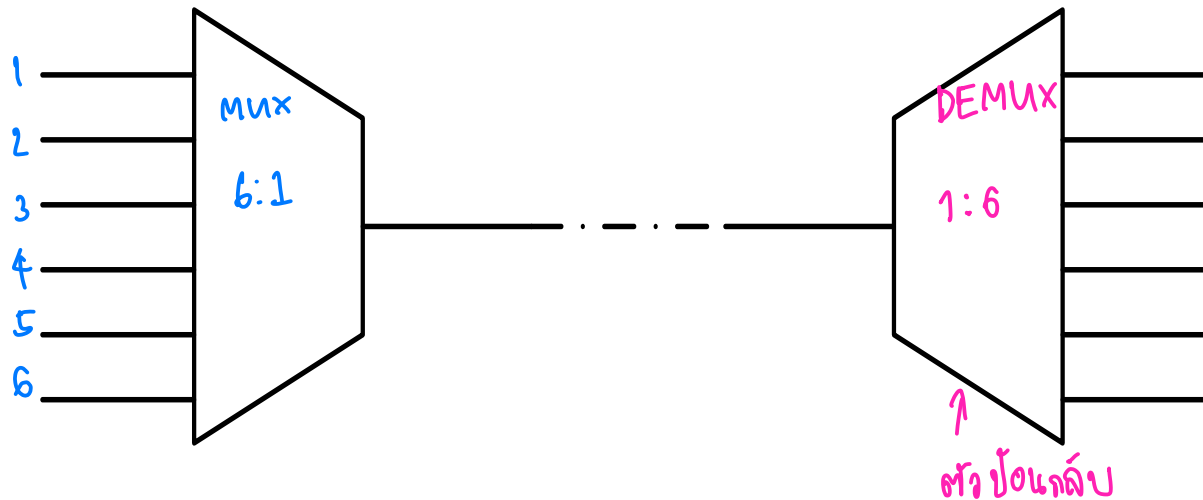
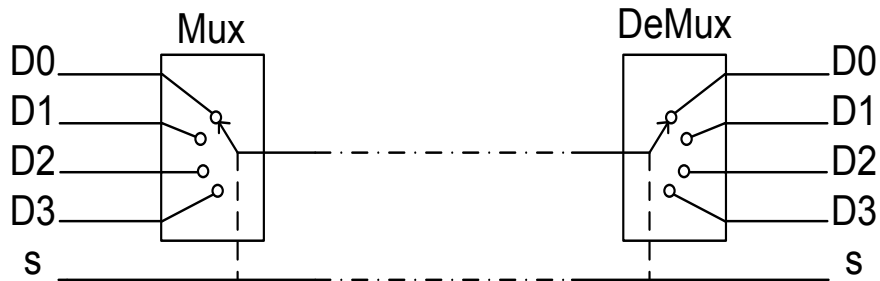
จำนวน



Compare between No MUX and MUX



What is circuit inside MUX and DEMUX



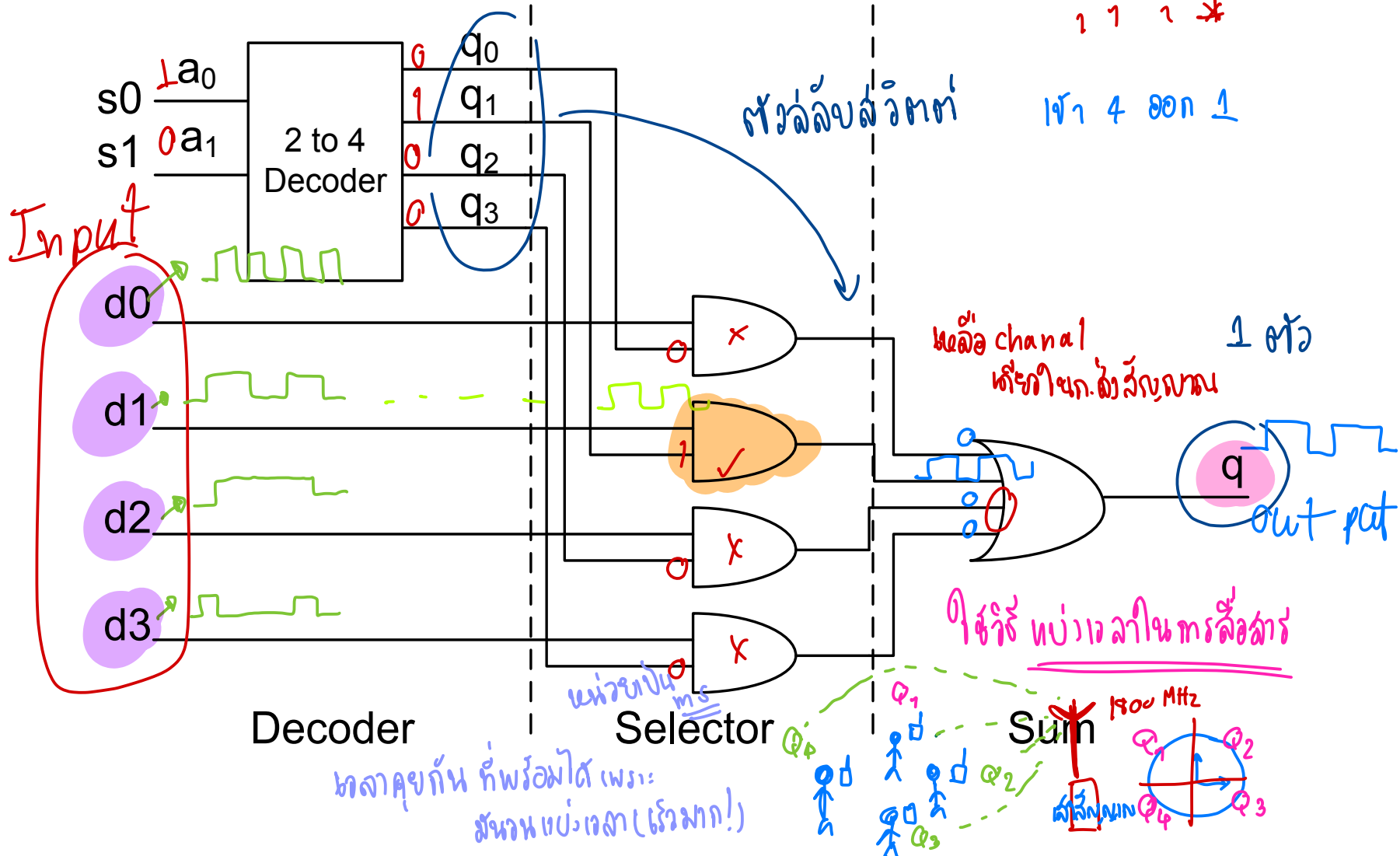
ดูจากภาพ

Multiplexer circuit

AND

0	0	0
0	1	0
1	0	0
1	1	1 *

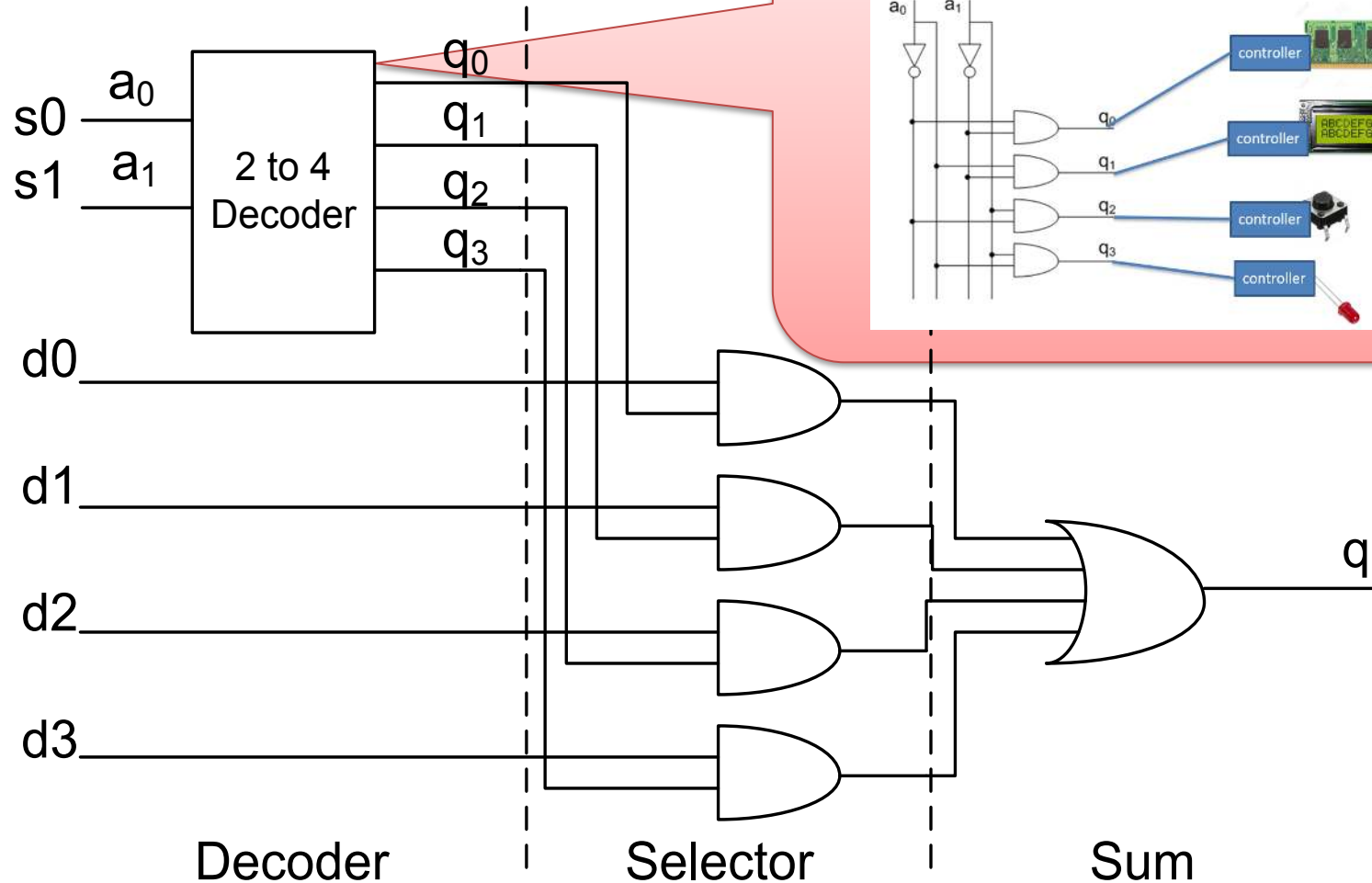
เข้า 4 ออก 1



หลายคู่กัน ที่พร้อมได้ เพราะ
สายนแบ่งเวลา (เร็วมาก!)

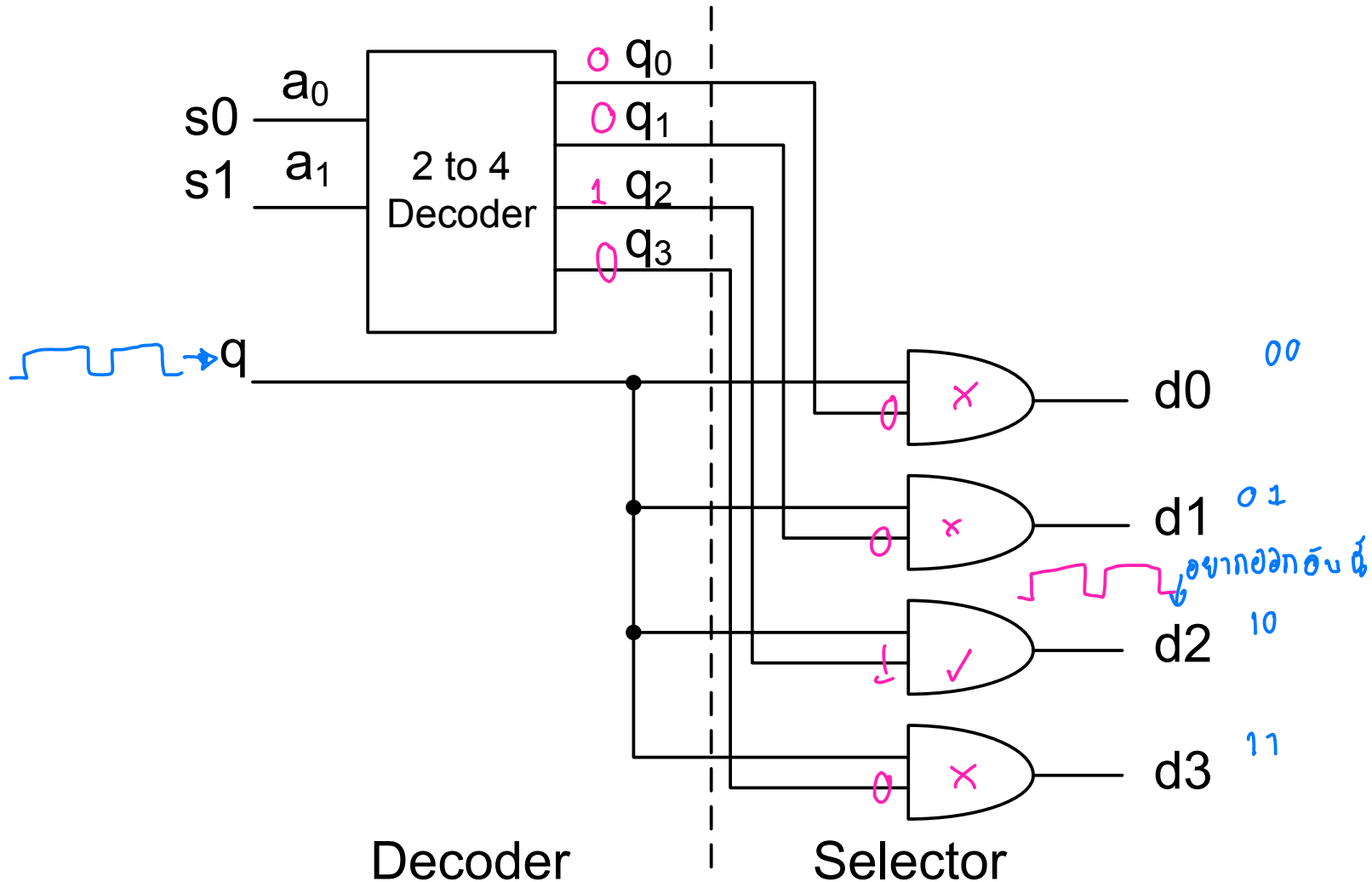
Multiplexer circuit *row 2*

Decoder

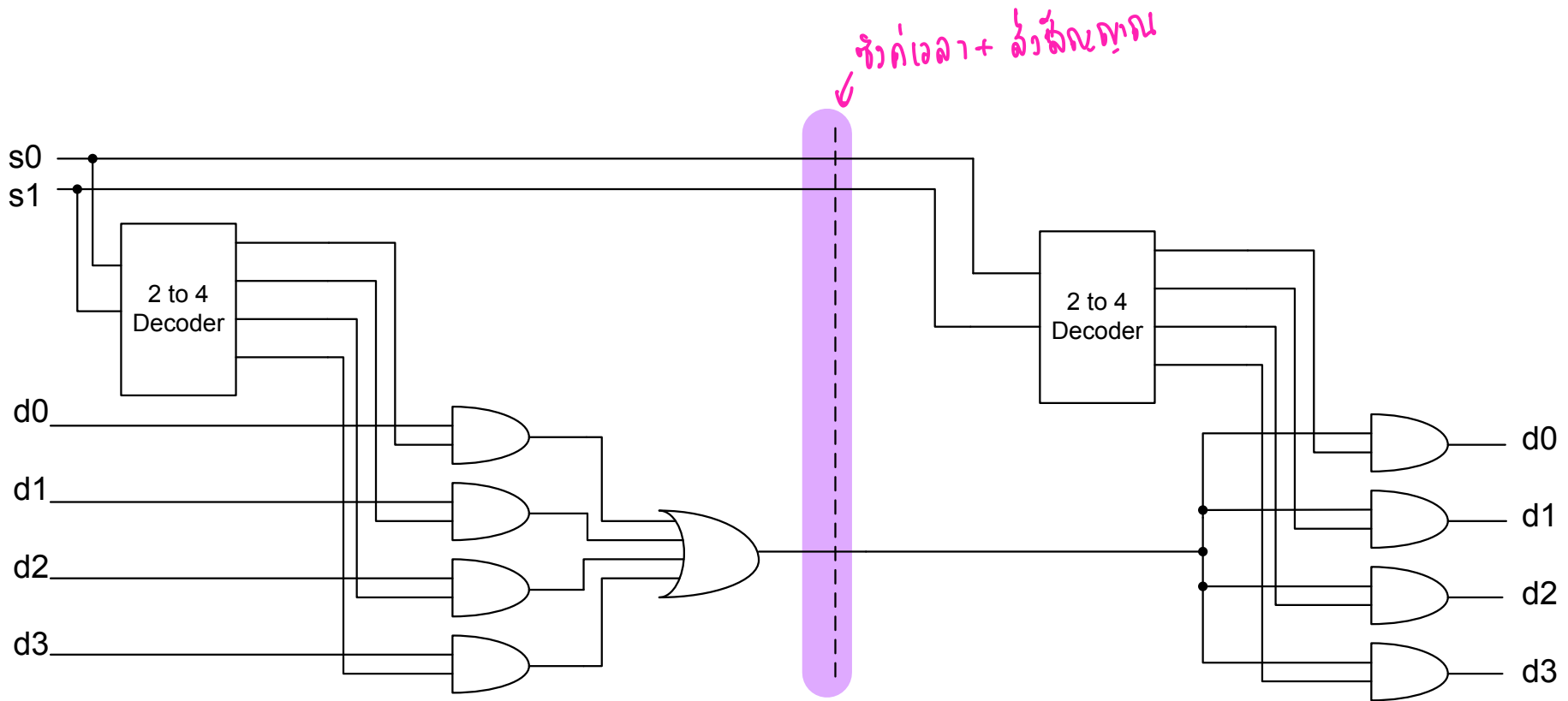


อยู่ในโทรศัพท์มือถือ! OMG!!

De-multiplexer circuit



MUX + DEMUX



Example devices use MUX/DMUX

จริงๆ แบ่งเวลาใช้แต่แบ่งรับเวลาเร็วขึ้น

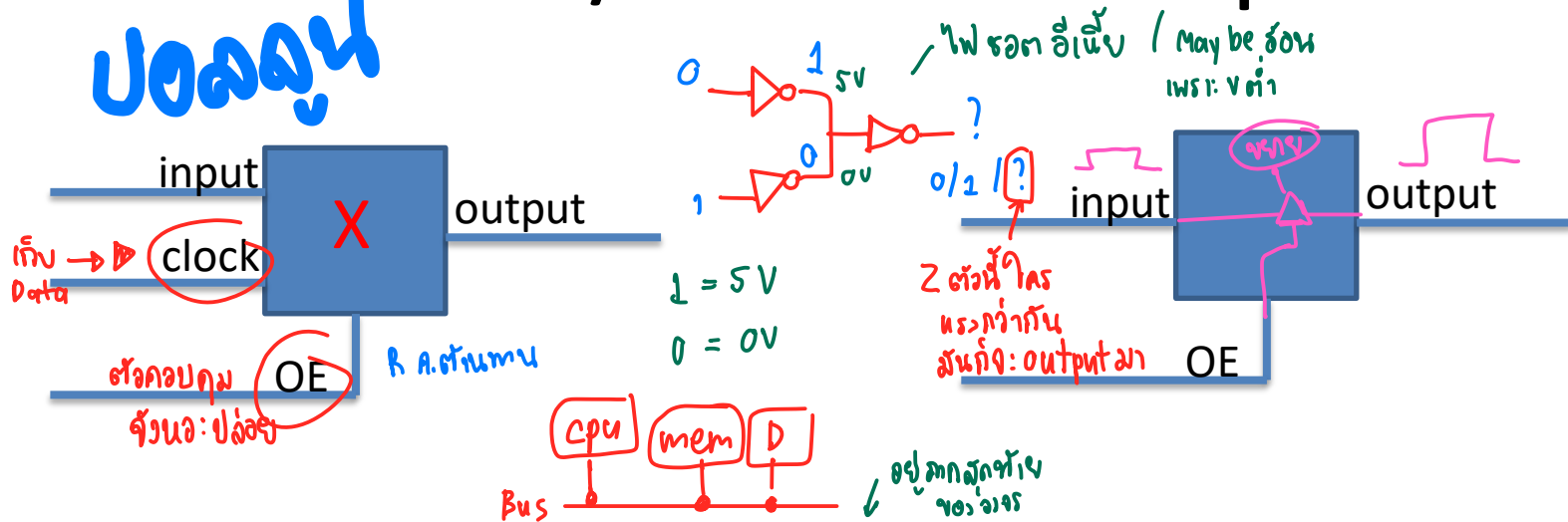
เลือกเส้นเดียว

จากเส้นเดียว
เลือกออกหลายเส้น



คล้าย
ปอดดูน

Latch/buffer concept



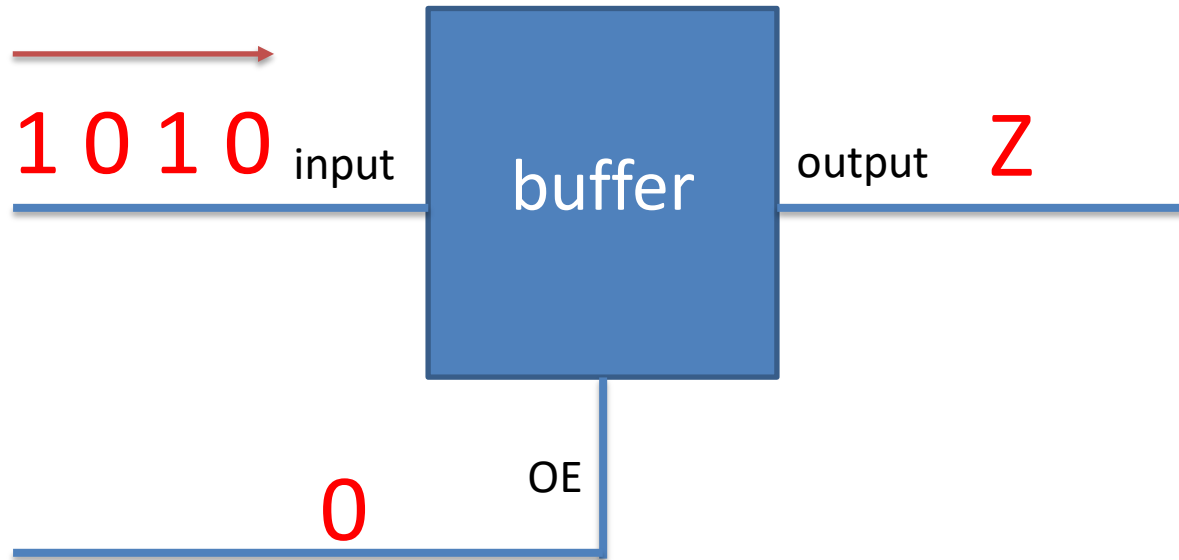
Latch circuit functions

- Hold data from inputs
- Control the data holding with clock/enable pin
- Have an output enable pin to control on/off output signal

Buffer circuit functions

- No hold logic data
- Pass through by amplify the signal
- Some buffer has an enable pin to on/off output signal

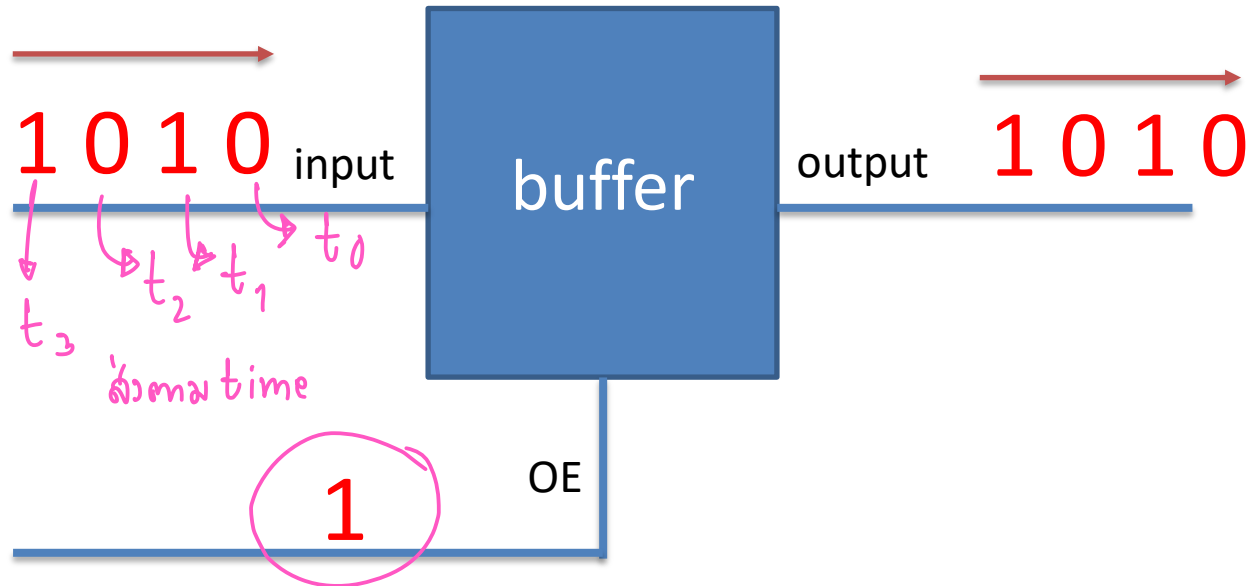
Work of the buffer circuit



ไม่ใช่ output อะไรออกมา

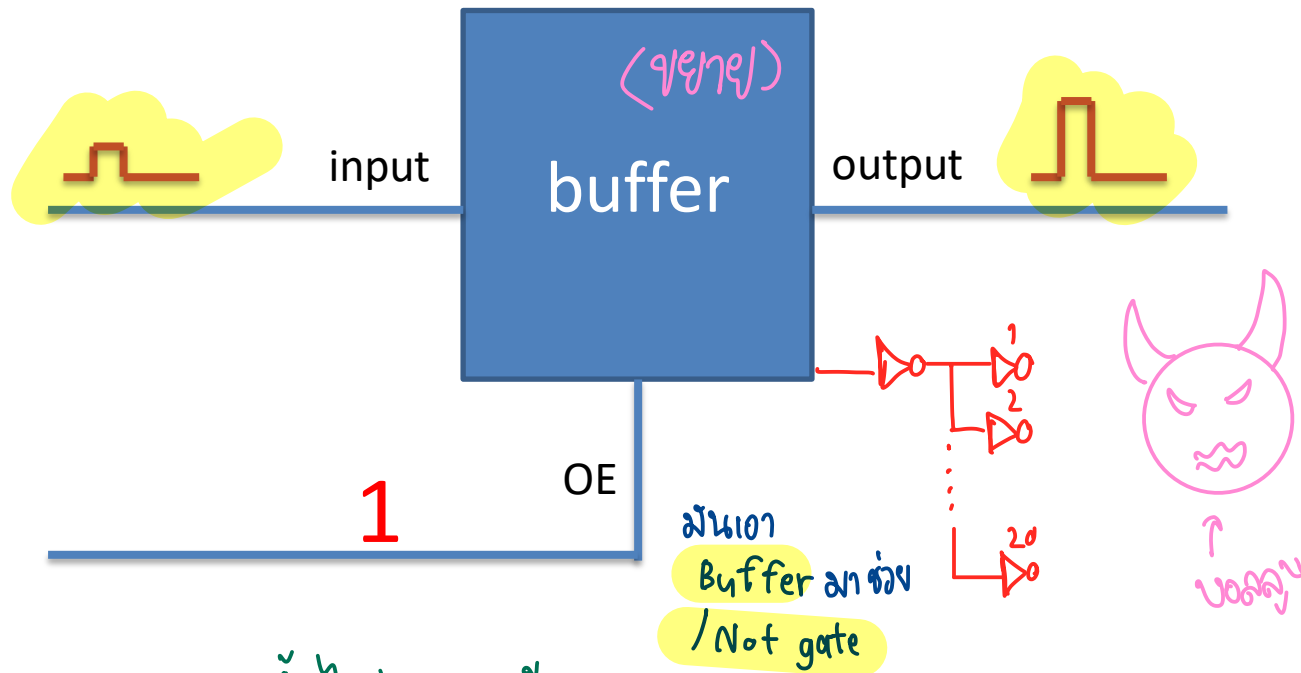
“Z” is a high-impedance state similarly no-connection state.

Work of the buffer circuit



“Z” is a high-impedance state similarly no-connection state.

Work of the buffer circuit



รูปนี้ ไม่ ขอตจร้า

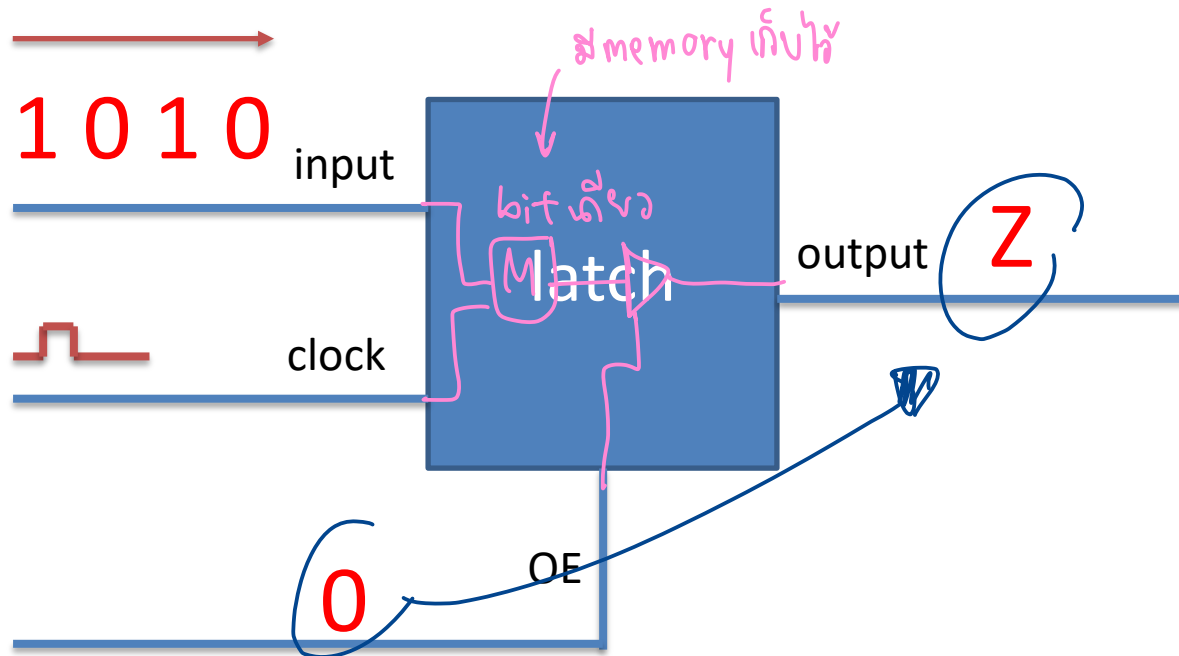
ตอนนี้ขอตจร้าไม่ OK ไฟไม่พอ
↑ ต่อหน้า ๗๐.1 ลักษณะ

เป็น ๗๐.๑ ผ่านเองๆ

“Z” is a high-impedance state similarly no-connection state.

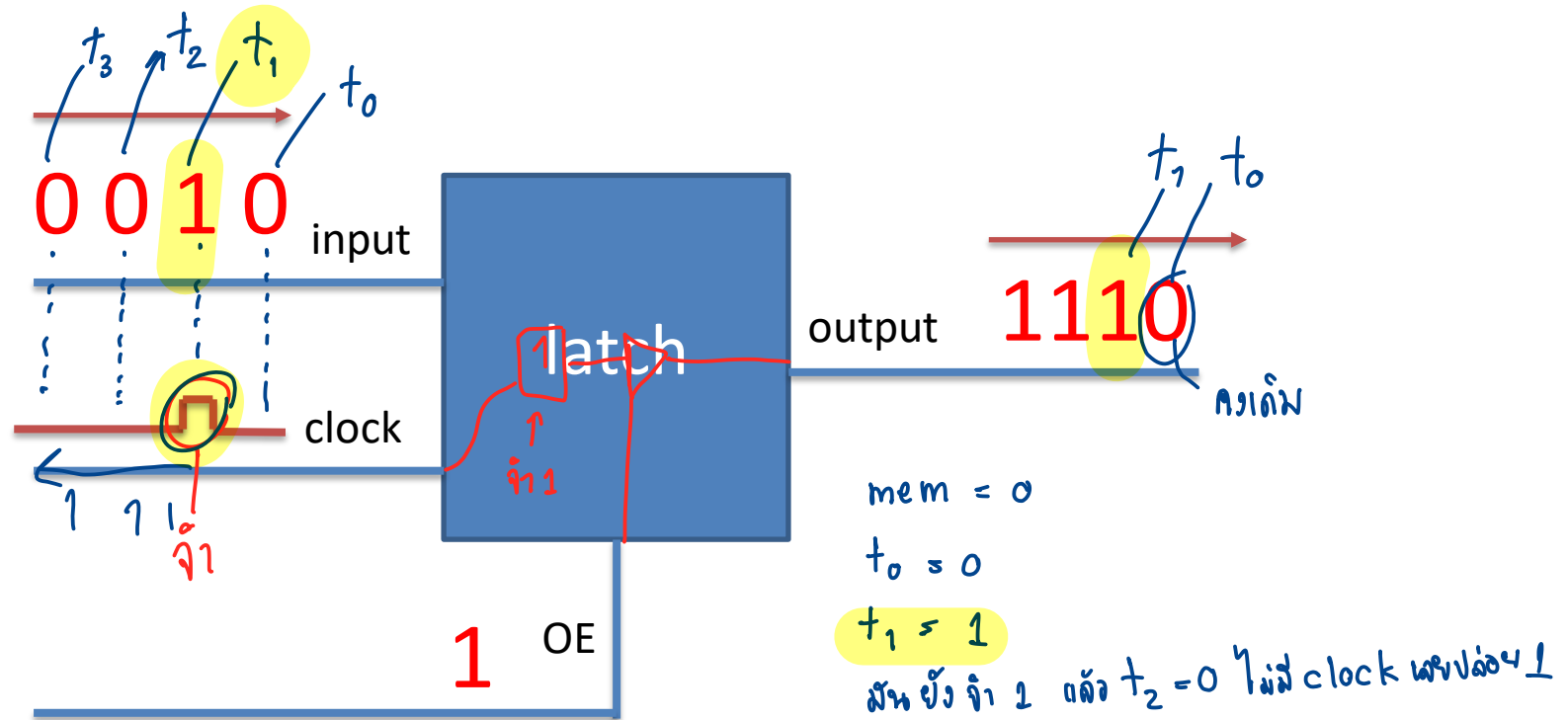
Work of the latch circuit

เข้ามามีทั้งใน mem แล้วส่งผ่าน



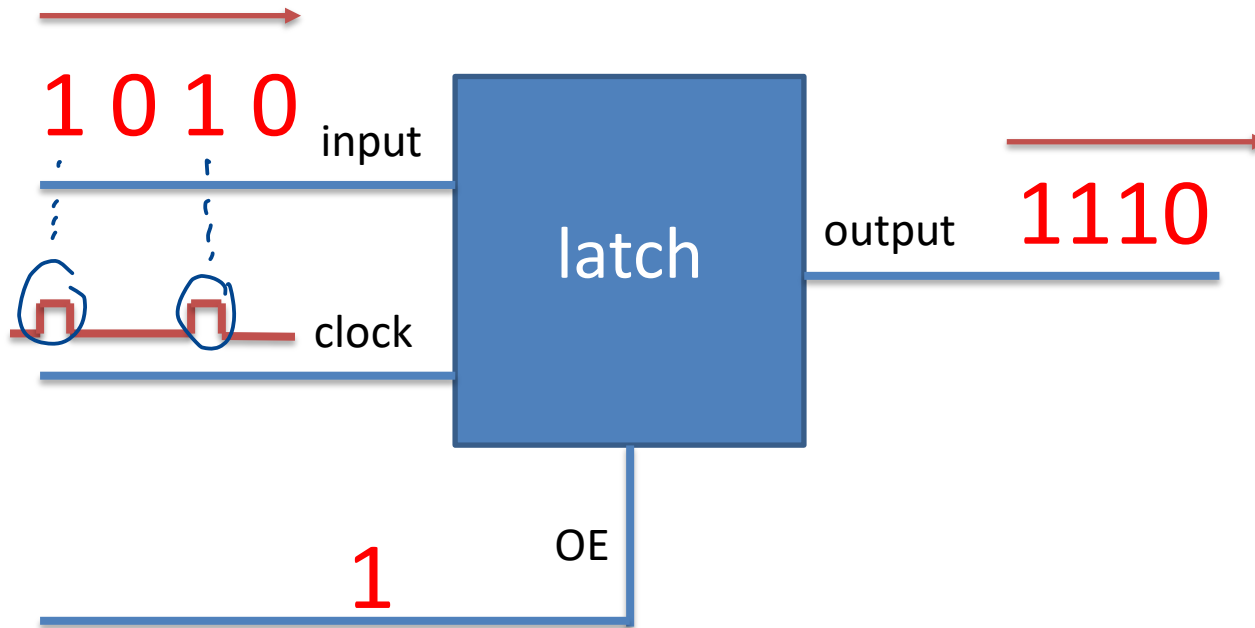
“Z” is a high-impedance state similarly no-connection state.

Work of the latch circuit



“Z” is a high-impedance state similarly no-connection state.

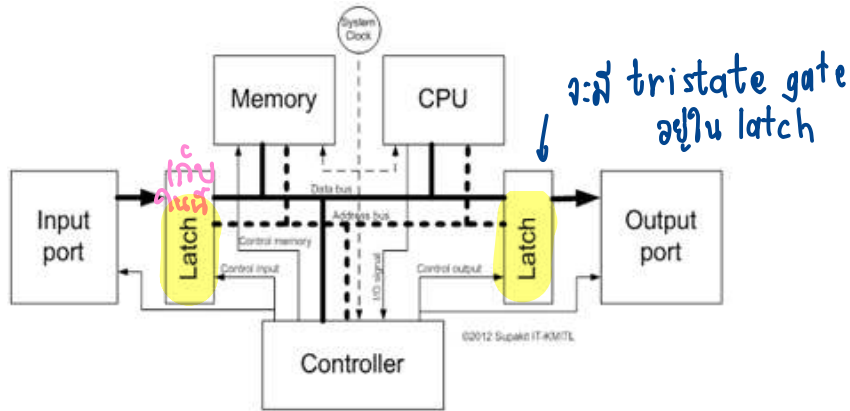
Work of the latch circuit



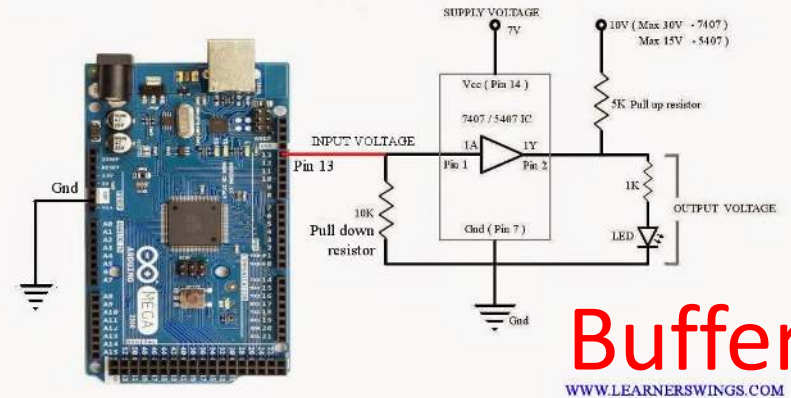
“Z” is a high-impedance state similarly no-connection state.

Latch/buffer in the computer

ຈອດ ຈຳນວນ



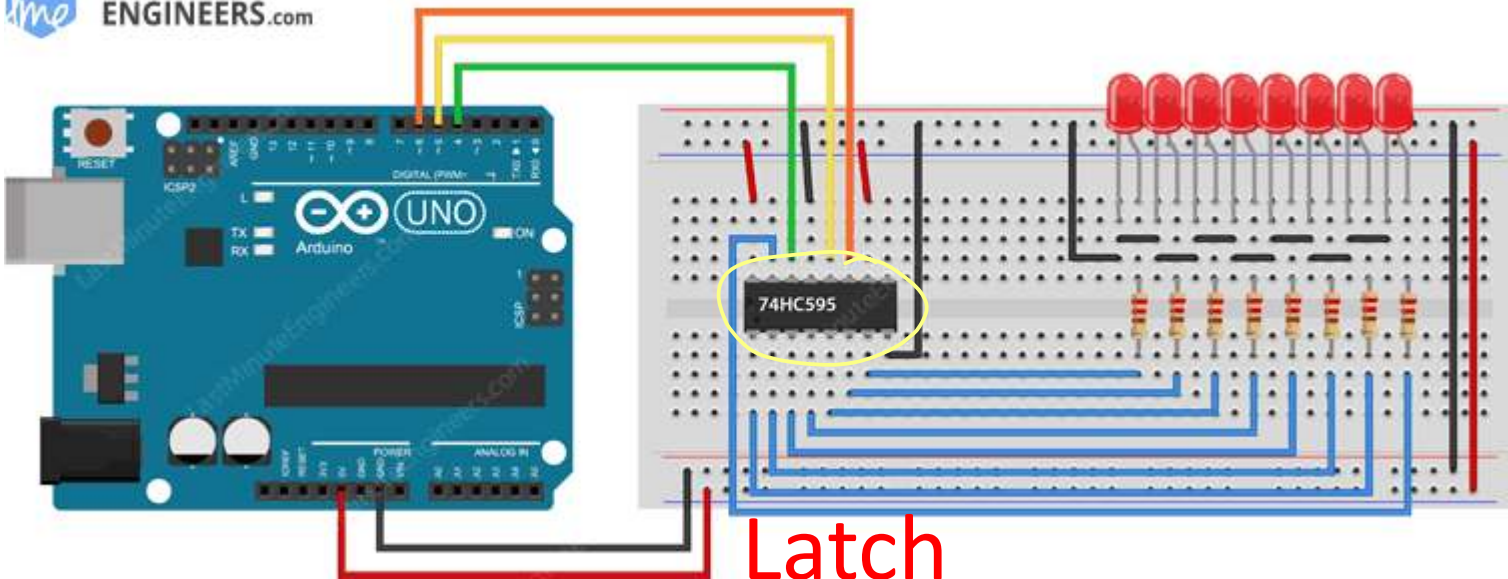
WWW.LEARNERSWINGS.COM



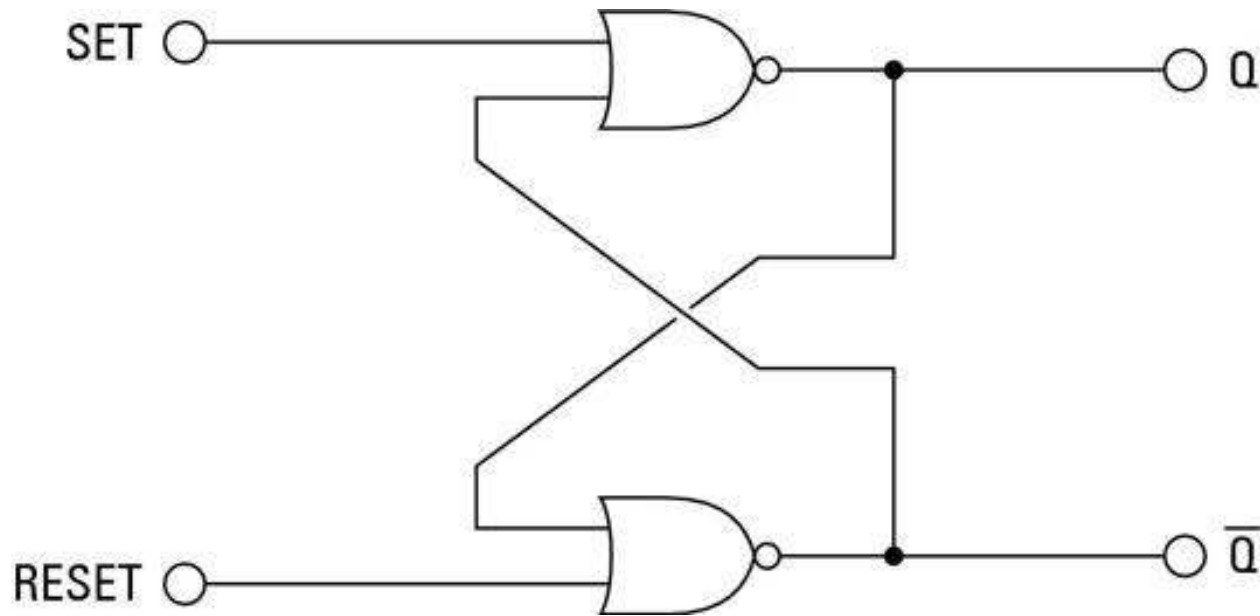
Buffer

WWW.LEARNERSWINGS.COM

Last Minute ENGINEERS.com



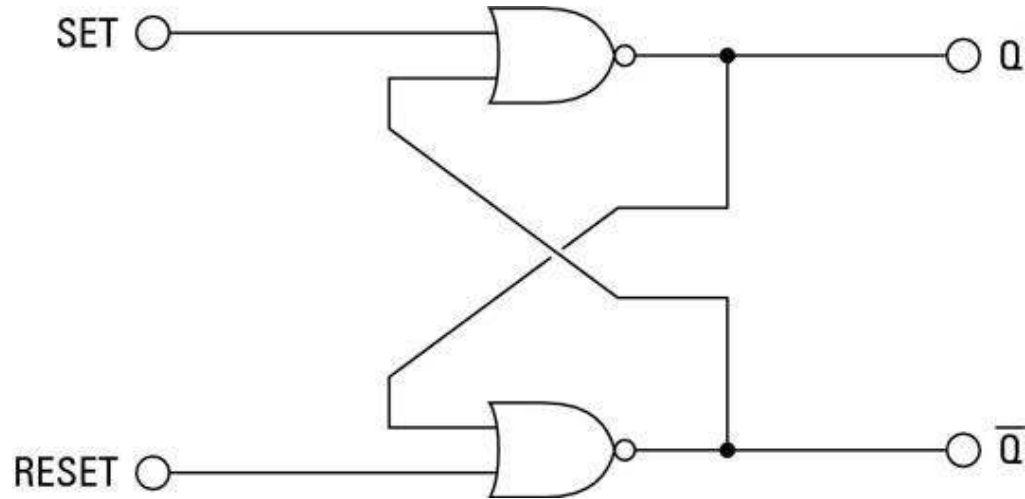
What is a digital circuit inside the latch?



lab 4 อย่างต่อเส้นทวน

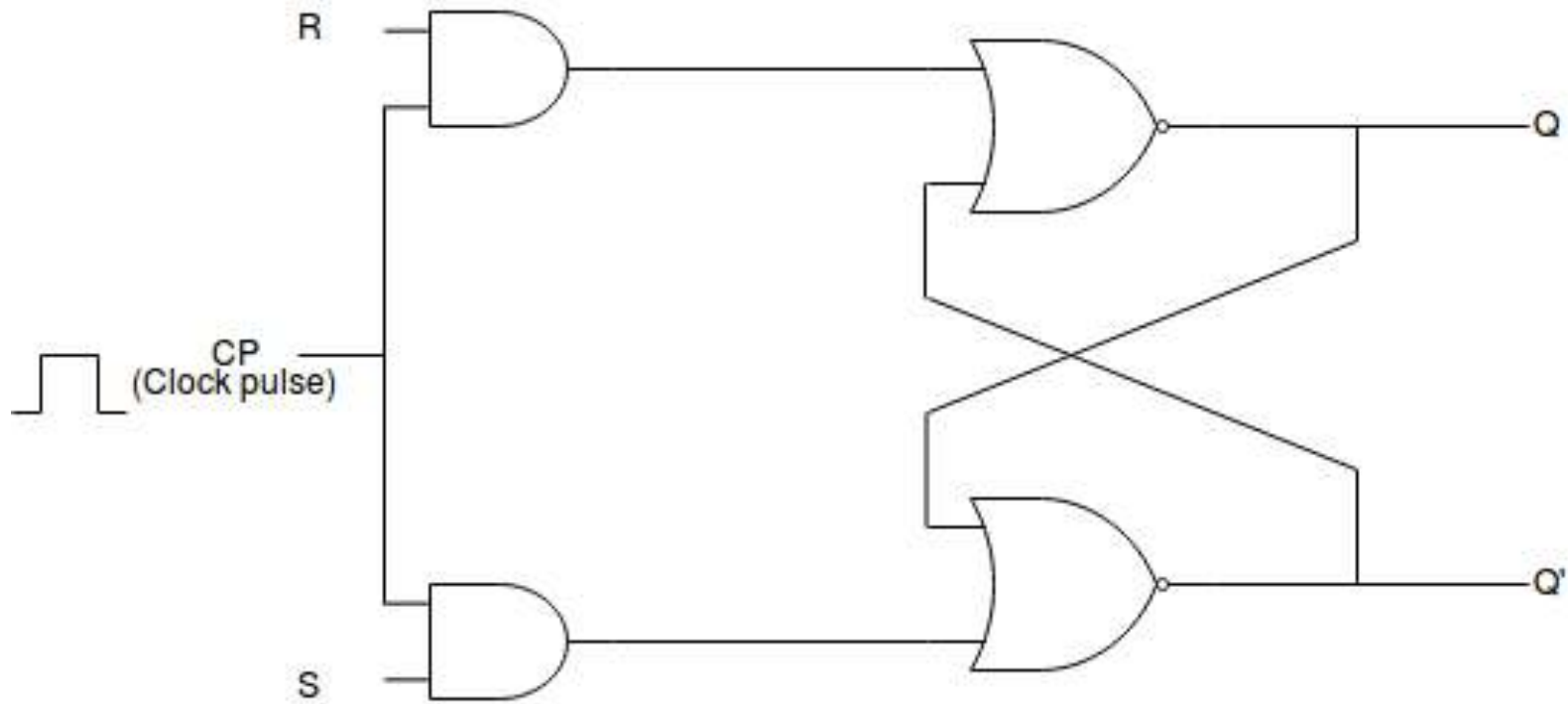
SET/RESET Flip flop = SR Flip-flop

Activity 3.1 Write truth table of SET/RESET – Flip Flop



SET	RESET	Q	Q'
x	x	0	1

SR-FF with Clock



Reference

นพ. นั้ จดเอนกัไว้เป็น
อีกมา๗๗

พ.ง.ฟองน้ำสี